An independent exact recheck of the classification of bistable two-species mass-action networks

(a recheck of a result by others; no novelty is claimed)

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AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was audited by a separate AI agent instance (`AUDIT.md`, ACCEPT_WITH_CORRECTIONS), not by a human.** The five corrections are applied (Section 6). No human audit exists. There is no Lean in this paper. The subject is pure mathematical dynamical systems; no biological or medical claim is made.

Abstract

B. Peng [1] (arXiv:2610.32739, 23 Oct 2026, unrefereed) classifies the bistable networks among small two-species bimolecular mass-action networks with at most three directed reactions: 681 orbits under species exchange, of which 27 are bistable (two positive equilibria that are sinks), with rational witnesses (his Table 2) and 654 exclusions (JAC 245, DIV 216, ROOT 179, FLUX 14). **The result is Peng's; we claim nothing new.** Two independent implementations reproduce the enumeration (1351 labelled edge sets, 681 orbits), verify all 27 witnesses exactly (each has exactly three positive equilibria: two sinks and a saddle), and reproduce all four exclusion counts; the ROOT count 179 is reproduced only by the auditor, with the resultant taken at the actual degree of each polynomial (the producer's fixed-degree reading gave 105). The exclusions are certificate checks conditional on the paper's Lemma 3 (pen-and-paper, not machine-checked); Theorem 2 is not verified. Random searches (Tier [X]) found no bistable network among the 654 excluded ones. Outcome: PARTIAL. Tier [C] for the exact parts, [X] for searches, [L] for the paper's lemmas.

What this is NOT. Not a new result; not a proof that the 654 networks are non-bistable independent of Peng's Lemma 3; not a check of Lemma 3, its Sard step, the JAC midpoint identity or Theorem 2; not a formal proof (no Lean); not based on the author's code or catalogue; not an exhaustive prior-art search; not human-audited; no biological or medical claim.

Markers. [L]: a cited source or lemma, taken as stated. [C]: executed exact code (rational arithmetic, symbolic algebra, certified ball arithmetic) with the argument written down, not formalised. [X]: searches and context controls, not exhaustive. Identifiers `BIST-··-nnnn` refer to `ledger/ledger.json`.

1 The class and the enumeration ([C])

Following Peng, the complexes are $X, Y, 2X, X+Y, 2Y$ and a network is a set of at most three distinct directed edges $\alpha \to \beta$ between them with $\alpha \neq \beta$ (20 possible edges). With mass action, $H = \sum_j k_j(\beta_j - \alpha_j)z^{\alpha_j}$ and the dynamics are $z' = (a,b) - z + H$ (model E) or $z' = (a,b) - \mathrm{diag}(\rho,1)z + H$ (model F), all of $a, b, k_j, \rho > 0$. The target ("bistable") is: two distinct positive equilibria with $\mathrm{tr}\, DF < 0$ and $\det DF > 0$ (sinks). The paper uses E with $\rho = 1$ for the witnesses.

Both implementations (own code each) find 1351 labelled edge sets, of which 11 are fixed by exchanging X and Y, giving 681 orbits of sizes 1/10/100/570. Numbering the orbits by first appearance with the complexes

and edge sets in lexicographic order reproduces all 27 identifiers of the paper's Table 1, and the 27 sets lie in 27 distinct orbits (`BIST-C-0001`). The paper does not state the representative rule; first appearance reproduces it. The paper is internally inconsistent on the label count: Sec. 1 gives 1351, Sec. 5 says "all 1362 labels" ($1362 = 1351 + 11$); we take 1351, which both implementations obtain.

2 The 27 witnesses ([C])

Producer: for each Table 2 parameter set, Gröbner elimination of y, square-free resultant, exact Sturm count of positive roots, $y > 0$ and the signs of trace and determinant by rational intervals of width 10^{-40} (gcd of the polynomial and the trace/determinant polynomial being 1). Auditor, by a different method: the exact resultant in y, its square-free part, certified root isolation with Arb ball arithmetic (python-flint, 400 bits), both y-roots of the second equation in ball arithmetic, candidates discarded when the first equation is certified nonzero. Result in both: *every one of the* 27 *has exactly three positive equilibria, two sinks and one saddle (*$\det < 0$*), with no undecided sign* (`BIST-C-0002`). The x-coordinates agree (e.g. N0438: 0.020611, 2.02, 397.909).

The Table 2 values were first read by the producer from flattened PDF text; four rows were mis-grouped and fixed by testing all groupings. The auditor then confirmed all 27 rows against the rendered PDF image (the author's catalogue was not used). Rows N0514, N0524: $k_1 = 17$, $k_2 = 100/9$; rows N0548, N0554: $k_1 = 17$, $k_2 = 100/159$; $k_3 = 2600/2809$, $650/729$, $2600/2809$, $2600/2809$ respectively.

3 The 654 exclusions ([C], conditional on [L])

The paper excludes bistability by four certificate families, each valid on the paper's Lemma 3 (projection injectivity, "at least three equilibria for two sinks", and a Sard argument for the saddle) and, for JAC, a midpoint identity. We read these arguments; they look correct, but they are pen-and-paper and neither of us machine-checked them ([L]). What we checked is that the certificates exist for the claimed networks.

Rule	paper	reproduced	by
JAC ($\det DF$ has nonnegative coefficients and the monomial ρ with coefficient 1)	245	245	producer, auditor
DIV ($\det K$ nonnegative, monomial ab with coefficient 1), after JAC	216	216	producer, auditor
ROOT (Sylvester resultant sign pattern)	179	179	auditor (producer: 105)
FLUX (copositivity on the flux cone)	14	14	auditor only

Table 1: Exclusion counts; total $245 + 216 + 179 + 14 = 654$. None of the 27 positives is excluded by any rule.

JAC and DIV. Applied in that order, both implementations obtain 245 and 216, split by network size (0 edges 1/0, 1 edge 8/2, 2 edges 54/33, 3 edges 182/181) exactly as in the paper's table; 193 negatives remain, including all 14 FLUX networks.

ROOT. The auditor takes the Sylvester resultant at the actual degree of each polynomial in the eliminated variable, eliminating x or y, with the paper's sign-pattern criterion (each coefficient fixed strict or free in $\{-1, 0, 1\}$, at most two sign variations, at least one fixed strict coefficient). This passes 179 of the 193: 13 with two edges and 166 with three, equal to the paper's per-size counts; 0 positives and 0 FLUX networks pass (eliminating y alone gives 171, x alone 126, union 179). The producer, reading "nominal degree" as a fixed degree-2 (4×4) Sylvester matrix, obtained only 105. The auditor reproduces 105 under that reading, so the gap of 88 comes from the producer's reading, not from the paper; the actual-degree reading is also sound, since a fixed Sylvester determinant vanishes at every common root even under degree drop and the strict coefficient excludes the zero polynomial.

FLUX. The auditor enumerated the rays of the flux cone $\{w \geq 0, U \geq 0\}$ (6 to 9 per network) and formed $G = R^\top H R$, checking $w^\top H w = \det M(w)$ symbolically. Eleven networks have G entrywise nonnegative; N0391 gives $u^2 - 2uv + 4v^2$ and N0646 gives $u^2 - uv + v^2$ (both positive definite); N0610 gives $u^2 + 3uv + 2v^2 - uw - vw + w^2$, copositive but not positive semidefinite, with the paper's decomposition $(w - \frac{u+v}{2})^2 + \frac{3}{4}u^2 + \frac{5}{2}uv + \frac{7}{4}v^2$ checked by hand expansion. So FLUX 14/14 reproduces (`BIST-C-0003`).

4 Controls ([C])

Positive: the Sec. 4 baseline ($Y \to 2X$ with $k = 10$, $2X \to X + Y$ with $k = 50/9$, $X + Y \to Y$ with $k = 10$; $a = 4/99$, $b = 1/90$) has exactly the equilibria $x = 1/10$, $1/5$, $3/5$ with det $= 25/9$, $-20/9$, $100/9$ (exact substitution) and trace formula as in the paper; as a is raised, two sinks at $a = 0.04610$ and one at $a = 0.04611$, consistent with the fold at $a_+ \approx 0.0461033$. Negative: the empty network, $X \to Y$, $X \leftrightarrow Y$ and $X + Y \leftrightarrow 2X$ each give exactly one positive equilibrium (a sink); the N0056 network ($X \to X + Y$, $X + Y \to 2X$) gives exactly one at the producer's tested parameters but can give none (e.g. $a = 1/3$, $b = 2/7$, $k = 3/2$, $h = 5/4$, $\rho = 1$), consistent with the paper's bound of at most two. Perturbation: scaling the inflow a by $2, 5, 20, 100, 1000$ destroys bistability for all 27 witnesses (`BIST-C-0004`).

5 Random search ([X])

Producer: 125,000 log-uniform samples of (a, b, ρ, k) in $[10^{-3}, 10^3]$ for each of the 681 orbits (85.1M samples), every hit re-verified exactly in rationals: all 27 positives re-found (hit rates 0.34% to 8.4%); 0 candidates among the 654 negatives. Auditor, independent pipeline: 20,000 samples for each of the 27 positives and 193 ROOT/FLUX negatives (4.4M): all 27 re-found (lowest rate 0.39%); the float pipeline flagged 247 samples on 10 negative networks, all rejected by exact Arb re-check (none with two sinks, none undecided). Hence no bistable negative was found. This is evidence only, limited by the parameter box and sampling, and the producer's "0 candidates" depends on its own pipeline (`BIST-X-0001`).

6 Audit corrections applied

The producer's `data/RESULT.md` and `data/result.json` are kept unmodified (pre-audit); this paper supersedes them on the audit's five corrections. (1) Outcome: the producer's run reproduced ROOT only partly (105) and FLUX not; the independent audit reproduced ROOT 179/179 (actual-degree resultant) and FLUX 14/14, conditional on Lemma 3; the status stays PARTIAL because Lemma 3 and Theorem 2 are not checked and the producer itself did not reproduce ROOT and FLUX. (2) The "88 not reproduced" remark is replaced by Section 3: it is the producer's fixed-degree reading, not the paper. (3) Table 2 is confirmed against the PDF image; the author's catalogue was not used. (4) The N0056 control reads "exactly one positive equilibrium at the tested parameters" (it can be zero). (5) The not-checked list is as in Section 7, with the paper's own 1351 versus "1362 labels" inconsistency noted in Section 1. The preregistration (`PREREG_ITEM.md`, sha256 `9c358006...4de3d3af`) is self-attested; it was written before computation according to the producer, with no independent timestamp.

7 Limitations and what is not done

- AI audit only; no human audit; no Lean (no Tier A).
- Not verified: Peng's Lemma 3 and its Sard step, the JAC midpoint identity (read, not machine-checked), Theorem 2 (deletion-minimality and inheritance), and the claim $B_E = B_F$ beyond the counts. The 654 exclusions are reproduced as certificate checks, not as a proof independent of Lemma 3.
- The author's repository, supplementary certificates and checker were not used (repository URL not in the paper; licence unchecked).
- Search limits: parameter box $[10^{-3}, 10^3]$, finite samples; model E only at $\rho = 1$ for the witnesses.

- Literature: no independent verification or critique of the paper and no earlier enumeration of this exact class was found, but the search was limited ([L], `BIST-L-0002`); related work includes Tang and Xu [2], not examined here beyond the abstract.
- Trusted components: sympy, python-flint (Arb), Python `fractions`.

8 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Producer code (`p1_enum.py`–`p7_aggregate.py`, `run_search.sh.txt`), summaries and logs are in `data/`; the auditor's scripts and small outputs are in `data/audit`. The 50 raw search part files remain in the lane directory only. The paper's full text is third-party and is not shipped. Nothing was committed, pushed or uploaded.

References

[1] B. Peng, *Bistability with equal and independent removal in small bimolecular networks*, arXiv:2610.32739 (v1, 23 Oct 2026), unrefereed. ([L])

[2] X. Tang, H. Xu, *Multistability of Small Reaction Networks*, arXiv:2008.03846 (v1 10 Aug 2020; v2 28 Jan 2021). ([L])